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A regional ANN-based model to estimate suspended sediment concentrations in ungauged heterogeneous basins

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ABSTRACT

The high cost of monitoring suspended sediment concentration (SSC) in rivers calls for the development of indirect estimation methods, based on relationships with other variables, which are easier and cheaper to measure. We present an original approach to investigate the capacity of regional models to extrapolate SSC to ungauged basins, in a heterogeneous region with scarce in situ data and complex hydrography. The estimates were based on qualitative variables (drainage area, soil type, land use, land cover and mean catchment slope) to represent spatial variability, and quantitative variables (turbidity, flow, precipitation and exponentially weighted moving average of past rainfall) to represent temporal variability. We used artificial neural network (ANN)-based models, applied to the Brazilian part of the Upper Paraguay River Basin, covering an area of 362 380 km^2. This study demonstrates that the proposed methodology allows the regional extrapolation of SSC to ungauged basins with very good performance, even in heterogeneous regions.

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1 Introduction

Suspended sediment concentration (SSC) is an important component of water quality that directly influences the biological, physical and chemical processes of water bodies. Monitoring this variable is important for studies of erosive processes in hydrographic basins, for planning and operation of reservoirs or hydro-electric plants and for the recovery of degraded aquatic environments.

The monitoring of SSC generally consists of field collection of water samples in previously determined sections and subsequent laboratory analysis. Due to high costs, technical constraints and difficulties in accessing remote sites, the sampling frequency and spatial coverage are usually low, and the spatio-temporal representation of the sediment dynamics is limited (Bayram *et al.* 2012, 2014, Sari *et al.* 2017, Fagundes *et al.* 2020). The lack of information about SSC in rivers has stimulated the development of indirect estimation methodologies based on the relation between sediment and other variables that are easier to determine over time.

Several approaches for estimating sediment transport have been developed over the last few decades, from the traditional sediment rating curve, a logarithmic regression that relates the liquid discharge to the solid discharge, to sophisticated data-driven approaches involving different types of input variables. Artificial neural network (ANN) is a common and well-known data-driven approach that lately has found applications in a variety of areas, including sediment estimations (Afan *et al.* 2016). Its high ability to model nonlinear and complex relationships is widely known and accepted around the world.

Discharge, previous suspended sediment load and precipitation (since precipitation is the main agent in hydrosedimentological processes) are the main input variables used in sediment estimation. Generally, these studies use long-term data series from single or a few hydrometric stations, producing satisfactory results (Jain 2001, Tayfur 2002, Kisi *et al.* 2012, Lafdani *et al.* 2013, Singh *et al.* 2013, Pektas and Cigizoglu 2017, Khosravi *et al.* 2018, Adnan *et al.* 2019, Nhu *et al.* 2020, Salih *et al.* 2020). However, most rivers around the world do not have a long-term data series of SSC for such applications.

Despite the widespread use of sophisticated data-driven approaches in sediment estimation, few studies have addressed ungauged basins and applications that deal with data scarcity. Heng and Suetsugi (2013) used ANN to estimate monthly suspended sediment load at ungauged sites of the Lower Mekong basin using climatic and flow data from seven hydrometric stations. Atieh *et al.* (2015) applied ANN models to estimate sediment transport in ungauged basins of Ontario, Canada. The authors integrated data from 94 basins to estimate the parameters of the traditional sediment rating curve using input variables that physically characterize the basins, as well as the climatic and hydrometric parameters, leading to satisfactory results. Although the applications involved ungauged sites, these studies used hydrometric stations with long-term data series to calibrate the models.

In recent decades, the use of turbidity as an indirect measure of SSC has gained prominence in programmes monitoring sediment transport in rivers due to high correlation with SSC, and because turbidity measurement is less expensive than traditional sediment

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collection (Bayram *et al.* 2012, Harrington and Harrington 2013, Shellenbarger *et al.* 2013, Haimann *et al.* 2014, Tananaev and Debolskiy 2014, Sari *et al.* 2015, 2017). Bayram *et al.* (2012) estimated SSC from turbidity data from six stations (144 records including all stations) along the Harsit River in Turkey using ANN models. Sari *et al.* (2017) developed an ANN model that related SSC data to turbidity and water level from the station installed in the basin of the Taboão River. Even with a small amount of data (59 SSC records), the ANN models yielded satisfactory results. The high correlation between turbidity and SSC allows satisfactory estimates even with limited data.

To deal with data scarcity, Teixeira *et al.* (2020) developed regional models based on ANN and a fuzzy inference system (FIS) to estimate SSC and turbidity using discharge, watershed area and hourly precipitation aggregated through the exponentially weighted moving average (EWMA) of past rainfall as predictive variables. The data monitoring was performed in four homogeneous watersheds within the same region in southern Brazil. Several FIS and ANN models performed very well for SSC prediction due to the inclusion of the EWMA variable. The results demonstrated that it is possible to estimate with little data for individual basins using a regional empirical model, as long as the basins have similar physical and climatic characteristics. In addition to the restriction regarding physically homogeneous regions, the capacity of the model for extrapolation to ungauged basins (basins that were not used in the training of the models) was not tested in that work.

The objective of this study is to investigate the capacity of regional models to extrapolate SSC to ungauged basins in a heterogeneous region with scarce in situ data, using qualitative variables to represent spatial variability and quantitative variables to represent temporal variability in each basin, thus filling important gaps in previous studies. ANN modelling was chosen to undertake this research, since it is one of the most frequently used and well-known data-driven approaches due to its simplicity and its ability to approximate any nonlinear relationship (Hornik *et al.* 1989).

The study was conducted in the Upper Paraguay River Basin (UPRB), Brazil, where the Pantanal biome, the largest tropical wetland area in the world, is located. This region is largely lacking in hydro-sedimentological monitoring, with approximately one station per 8000 km^2, and experiences low collection frequency (average of four records per year). The development of sediment rating curves, and of estimates from only hydrometric and climatic data, is unreliable due to insufficient SSC data. Turbidity is spatially and temporally monitored to a larger extent than SSC in the UPRB, allowing the use of this variable in SSC estimates.

The heterogeneity within the basin requires the use of qualitative variables that physically characterize the watersheds, such as soil type, land use, drainage area and mean catchment slope, as well as quantitative variables such as turbidity, discharge and previous rainfall. Due to the low density of precipitation gauges in this region, another novelty of this research is the use of a satellite rainfall product combined with the EWMA of past rainfall to aggregate temporal data and capture the dynamics of sediment formation from rainfall, even in ungauged regions.

2 ANN models

ANN models are among the most used and well-known data-driven models due to their simplicity and their capacity to approximate any nonlinear relationship. ANN models were inspired by the human brain and are composed of artificial neurons, which receive, modify and transmit signals, analogous to natural networks (Asce 2000). A feedforward neural network with a three-layer architecture (Fig. 1) is considered capable of approximating any continuous function to any designed degree of accuracy, provided there are enough neurons in the inner layer (Hornik *et al.* 1989).

In the first layer, the input data (explanatory variables) are inserted. In successive layers, each neuron receives the sum of the inputs, weighted by coefficients called synaptic weights. This sum, added to a bias term, goes through a linear or a nonlinear transformation (neuron activation function), which transforms the sum into an output value for this layer, constituting the output of the network (explained variables). During training, the synaptic weights of the connections and the bias are adjusted, from randomly chosen initial values (Freeman and Skapura 1991), by a learning rule, whose goal is to minimize

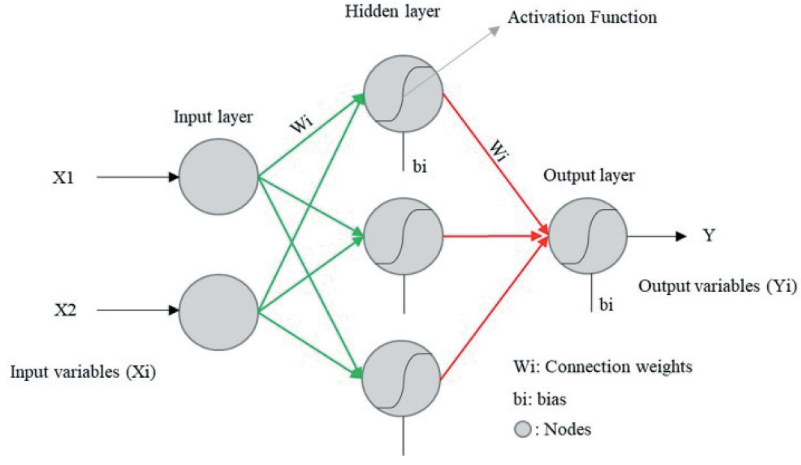


Figure 1. Representation of a feedforward artificial neural network (ANN) with three layers.

some measure of performance, based on the errors between the calculated and observed outputs (Asce 2000).

The back-propagation algorithm (Rumelhart *et al.* 1986) is one of the most widely used training methods for ANN. However, because of their high approximation capabilities, ANN models may be overly adjusted to the data, making them inappropriate for future applications. To prevent this overfitting, the cross-validation method (Hecht-Nielsen 1990) is usually employed, in which the available dataset is divided into three parts: training, validation and verification.

The cross-validation technique consists of separating out an additional training sample (validation sampling) to continuously evaluate the performance of the model during training. The onset of overfitting occurs in the cycle where the error in the validation series begins to increase while the error in the training series continues to decrease. Thus, cross-validation acts as a determinant of the ideal stopping point of the training, since training error statistics are always decreasing (indicating better performance), while an increase in the error statistic in the validation series indicates that a threshold has been reached.

The number of neurons in the inner layer is usually determined by trials. Sari *et al.* (2017) articulates a concept of parsimony for neuron number estimation. From an oversized network, the number of neurons in the inner layer is progressively decreased until a reduction in its generalization capacity, due to the lack of degrees of freedom, is observed. The smallest number of neurons observed before the reduction of the generalization capacity is chosen. The assessment of this condition must be carried out with the validation sampling since the verification sampling must not participate in the training of synaptic weights or in the ANN architecture decision (Hecht-Nielsen 1990). Finally, the trained network is tested on a verification sample that was not included in the training stage, to guarantee the generalization capacity of the model (Hecht-Nielsen 1990).

3 Study area and data

3.1 Pantanal wetland

This study was carried out on the Brazilian part of the UPRB, covering an area of 362 380 km^2, where the Pantanal, one of the largest wetland areas on the planet, is situated. The region (Fig. 2)

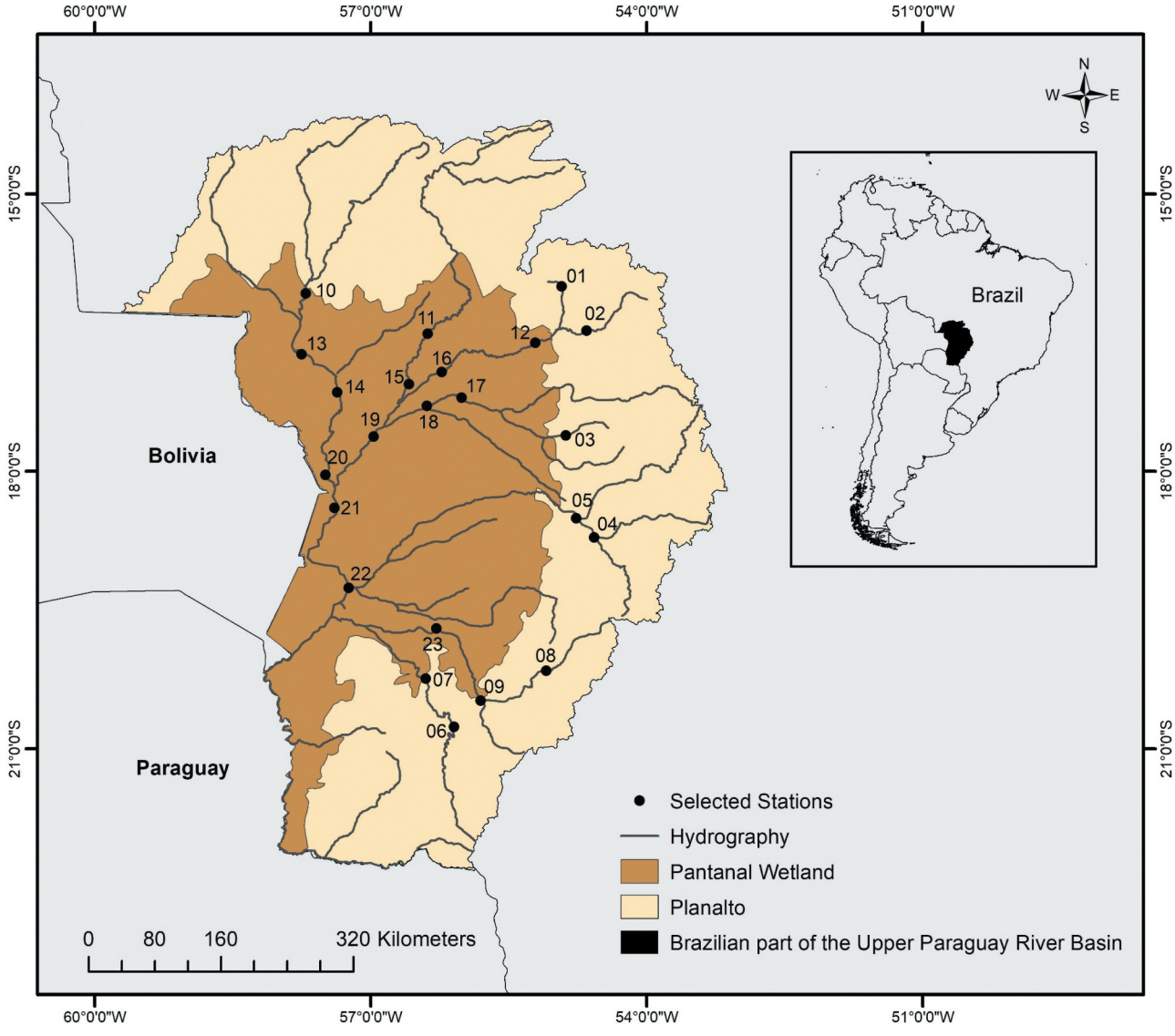


Figure 2. Localization of the study area, showing the selected hydrometric stations.

is subdivided into two areas according to distinct topographic and hydrological characteristics: the high-altitude Planalto region, and the lower Pantanal wetland region. Pantanal is also the name given to the biome of the plain.

The Planalto region corresponds to 59% of the basin area, with land at altitudes from 200 m to more than 1100 m, mainly in the eastern and northern parts. This part of the basin presents deep soils and high slopes, favouring rapid drainage and high sediment production.

The Pantanal plain corresponds to 41% of the basin area. This region presents low, flat and humid land with altitude below 200 m. The gentle slope and the low river banks make the drainage system unable to quickly transmit the water received from the Planalto, and extensive areas are flooded, resulting in complex hydrological and hydrosedimentological dynamics (Bravo *et al.* 2012).

According to the Köppen-Geiger climate classification, the predominant climate in the UPRB is tropical savanna, with rainfall concentrated in the summer. The annual average precipitation in the UPRB varies from 1200 to 1600 mm, with the highest values occurring in the Planalto. About 80% of the annual rainfall occurs in the rainy season, which runs from October to April (Bravo *et al.* 2012).

3.2 Data

For this study, 23 stations with hydrometric data were selected for model construction due to the coincident availability of SSC, turbidity and flow data. The data were provided by the Brazilian National Water Agency (ANA), through the national hydrological information system (Hidroweb) (www.hidroweb.ana.gov.br).

Information on the 23 stations is given in Table 1. The period of record starts in 2009, the year in which ANA initiated a water quality monitoring programme in the UPRB. Most of the sediment data collection was performed during the rainy season (October to April).

Table 1. Selected hydrometric stations.

	Station Name	River	Region	Drainage Area (km^2)	Period of Record
1	São Pedro da Cipa	São Lourenço	Planalto	3 267	2009-2018
2	Rondonópolis	Vermelho	Planalto	12 297	2009-2017
3	Estrada BR-163	Correntes	Planalto	3 969	2009-2014
4	Garimpagem	Jauru	Planalto	6 328	2009-2016
5	Coxim	Taquari	Planalto	27 573	2009-2016
6	Estrada MT-738	Miranda	Planalto	11 552	2009-2016
7	Miranda	Miranda	Planalto	15 189	2009-2016
8	Ponte do Grego	Aquidauana	Planalto	6 318	2009-2018
9	Aquidauana	Aquidauana	Planalto	15 707	2009-2016
10	Cáceres	Paraguay	Pantanal	32 373	2010-2016
11	Porto Cercado	Cuiabá	Pantanal	37 024	2009-2018
12	Acima do Córego Grande	São Lourenço	Pantanal	23 105	2009-2018
13	Descalvados	Paraguay	Pantanal	48 179	2010-2016
14	Porto Conceição	Paraguay	Pantanal	65 879	2010-2016
15	Ilha Camargo	Cuiabá	Pantanal	39 006	2009-2017
16	São José do Boriréu	São Lourenço	Pantanal	24 844	2009-2017
17	São Jerônimo	Piquiri	Pantanal	23 315	2009-2017
18	São José do Piquiri	Piquiri	Pantanal	29 388	2009-2017
19	Porto do Alegre	Cuiabá	Pantanal	102 580	2009-2017
20	Amolar	Paraguay	Pantanal	72 642	2009-2013
21	São Francisco	Paraguay	Pantanal	185 113	2009-2018
22	Porto da Manga	Paraguay	Pantanal	261 480	2009-2016
23	Porto Ciriaco	Aquidauana	Pantanal	17 190	2009-2016

The elevation of UPRB was obtained from a digital elevation model (DEM) of the Shuttle Radar Topography Mission (SRTM), with a spatial resolution of 30 m, made available by the United States Geological Survey (USGS). The Arc Hydro Tool in ArcGIS 10.3.1 was used to extract drainage area and mean slope data for the selected stations.

The soil classes were extracted from the Brazilian soil type map (at a scale of 1:5 000 000), which follows the Brazilian soil classification system (SiBCS) (Santos *et al.* 2018). SiBCS was chosen because although it has similarities to other international taxonomic systems, it aggregates information specific to Brazilian tropical soils; this means, however, that direct transcription to other systems is not viable.

The land use was characterized using a map from the Brazilian Agricultural Research Corporation. The map was made using images from 2014 taken by the Landsat 8 satellite, with a spatial resolution of 30 m (Embrapa 2015).

Although the Brazilian part of the UPRB has 92 conventional precipitation gauges, which corresponds to 3938 km^2 per station, this density is considered low. According to the World Meteorological Organization (WMO 2008), the recommended density for plains is 575 km^2 per station. However, satellite rainfall products are becoming increasingly available and accessible in near-real time, with almost global coverage. Temporal and spatial resolution, as well as measurement accuracy, are continuously improving owing to steady advances in sensor technology and new methods for merging various data sources. Several high-resolution products have reached an operational stage, with a rapidly growing scientific community using these data to supplement or even replace raingauge observations (Gebremicael *et al.* 2017).

For these reasons, the precipitation data used in this study were obtained from the MERGE product, which is a combination of observed rainfall data and satellite information, made available to all of South America. The MERGE data come from the Brazilian National Institute of Space Research (CPTEC/INPE).

The MERGE product uses rainfall data estimated by remote sensing to fill in rainfall information in locations with insufficient in situ monitoring. A radius of influence is assigned to the observed data, and precipitation data estimated via satellite are used to fill in the missing information. The rainfall estimates were obtained from the Tropical Rainfall Measuring Mission – Multisatellite Precipitation Analysis (TRMM-TMPA) for the period 1998 to 2014, and by Global Multisatellite Retrieval Measurement – Integrated Multi-satellite Retrievals for Global Precipitation Measurement (GPM-IMERG) for 2015 to the present. Further details can be found in Rozante *et al.* (2010).

The daily data are available in grids of 20 km, whose precipitations are accumulated every 24 h. The MERGE product was downloaded for the period January 2008 to December 2017, which encompasses the data period of the SSC measurements. The data were extracted for each station, covering an area with a radius of 100 km. Subsequently, the data were interpolated to the stations using the inverse square distance method (Burrough and McDonnell 1998).

4 Model development

4.1 Input and output model variables

To develop the regional SSC estimation model, 346 SSC records from the 23 selected stations were used as the output variable. Data on turbidity, discharge and rainfall comprise the input quantitative variables, used to enable SSC estimates in the different rivers of the hydrographic region. Turbidity and flow data were obtained from the same hydrometric stations as the SSC data and refer to the same day. The rainfall data were extracted considering the same day as the SSC data, as well as previous days (rainfall data from days t, t – 1, t – 2, t – 3, t – 4, and t – 5).

The EWMA technique was proposed by Moore (1982) to represent soil moisture conditions in prediction models based on precipitation series. This resource assigns greater weight to more recent precipitations in the composition of the moving average, promoting a synthesis of recent precipitation information in the form of a representative value, resulting in more parsimonious models.

The EWMA is applied to precipitation time series with constant intervals, according to Equation (1):

$$EWMA_t = \sum_{i=0}^{\infty} w_i P_{t-i} \quad (1)$$

where $w_i = \alpha\,(1-\alpha)^i$; P_{t-i} is the precipitation at time $t - i$; and α is a constant, in units of time.

The constant α relates to a parameter called half-life (h):

$$\alpha = 1 - 0,5^{(1/h)} \quad (2)$$

Since the length of the weight series is undefined, it should be truncated from a minimum value towards the past. Therefore, it is more practical to use a recursive equation (Equation (3)):

$$EWMA_t = (1-\alpha)\ EWMA_{t-1} + \alpha\ P_t \quad (3)$$

The EWMA was applied to the precipitation data obtained from the MERGE product for the 23 points of analysis, considering the half-life parameters h = 1, 2, 5, 10 and 15 d. The half-life parameters were chosen from a preliminary analysis of linear correlations with SSC.

The transformation of turbidity readings into SSC depends on factors such as size, shape, colour, refractive index, density, and mineralogical composition of the sediment particles (Sari *et al.* 2015). Since the model data are taken from different parts of a very heterogeneous hydrographic region, the other attributes that are inserted (mainly qualitative variables) give the model the ability to differentiate these factors.

The variables used to represent the heterogeneity of the UPRB, which in this study we refer to as qualitative variables, comprise the drainage area (km^2), soil type (13 classes), land use and land cover (14 classes), and mean catchment slope (%), for the 23 selected stations. The soil type and land use/cover were included as proportions of the basin area. Each soil type, land use and land cover class was included in the regional ANN-based model as an input variable. Figure 3 presents a summary of the input variables assessed to estimate the SSC (output variable) in the rivers of the UPRB.

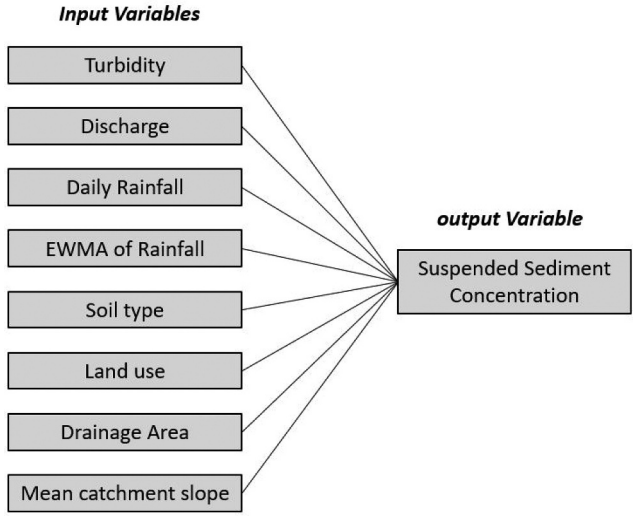


Figure 3. Summary of the input and output variables.

4.2 ANN models

Several ANN models were developed to estimate SSC from the reported input variables. From a complete model with all proposed input variables, other models were tested, leaving out one variable each time to determine its importance.

Multilayer perceptron models were used with a three-layer architecture, constituting an input layer, an intermediate layer and an output layer, all with a unipolar sigmoid activation function. The adoption of a single intermediate layer is based on the existence theorem (Hecht-Nielsen 1990), which states that an ANN with a single inner layer is capable of approximating any continuous numerical relationship. The back-propagation algorithm (Rumelhart *et al.* 1986) was custom programmed in the Matlab® R2012b environment. Overfitting was avoided using the cross-validation technique (Hecht-Nielsen 1990).

Regarding the number of neurons in the intermediate layer, we followed the concept of parsimony presented by (Sari *et al.* 2017), in which each ANN model was trained with cross-validation by reducing the number of neurons in the inner layer until a reduction in the performance of the validation sampling occurred. Then, the preceding alternative with sufficient degrees of freedom was chosen. This approach provides more parsimonious ANN models.

The approach adopted to divide the data sample for cross-validation was to use 11 stations for training, six stations for validation and six stations for verification (Table 2). Stations with extreme values (maximum and minimum) were included in the training samples. Stations from Planalto and Pantanal are present in all training, validation and verification samples, to provide spatial balance to the model.

Table 2. Sample separation.

Sample	Records	Station numbers
Training	166	1, 4, 5, 6, 8, 11, 16, 18, 20, 22 and 23
Validation	85	2, 9, 10, 14, 17 and 21
Verification	95	3, 7, 12, 13, 15, and 19

Each cross-validation training was performed with 15 repetitions, using random initiations of the synaptic weights. The training with the best final performance in the validation sampling was selected. This procedure was performed to minimize the influence of random initialization on the results.

4.3 Model performance statistics

The evaluation of the model performance was realized based on error statistics comparing the output from the network and the observed values. These error statistics were the mean absolute error – E_a (mean), the maximum absolute error – E_a (max), the median absolute error – E_a (50) and the 95th percentile of the absolute errors – E_a (95), and the Nash-Sutcliffe or NS efficiency coefficient (Nash and Sutcliffe 1970). The NS calculation is defined by:

$$NS = 1 - \frac{\sum_{i=1}^{N} (Ti - Yi)^2}{\sum_{i=1}^{N} (Ti - \bar{T})^2} \quad (4)$$

where T_i are the observed data; $\bar{T}$ is the mean of the observed data; and Y_i are the data calculated by the model. The NS ranges from $-\infty$ to 1; values near 1 represent good performance, whereas values $\leq$ zero indicate that $\bar{T}$ is a better approximation than the calculated values of Y_i.

To interpret the performance of the model via the NS, we used the classification proposed by Moriasi *et al.* (2015) specifically for sediments, in which the model is considered "very good" for NS > 0.8; "good" for 0.7 < NS ≤ 0.8; "satisfactory" for 0.45 < NS ≤ 0.7; and "not satisfactory" for NS ≤ 0.45.

5 Results and discussion

5.1 Data analysis

In this section, an analysis of all qualitative and quantitative data is performed, to provide a conceptual understanding of the results of the models. Figure 4 shows (a) the map of soil types, (b) the map of land use and (c) the DEM in the UPRB. The soil type and land use classes, as well as the drainage area and mean catchment slope for each hydrometric station, were obtained from these maps. In both maps it is possible to observe very different characteristics in the Planalto region versus the Pantanal region.

Regarding the soil type classes, the Argissolos, Neossolos and Latossolos are more frequent in the Planalto. This region presents soils with great depths and strong drainage, with colours ranging from reddish to yellowish and, more rarely, brownish or greyish. The most common soil classes in the Pantanal region are the Gleissolos, Espodossolos and Planossolos. These soil types have poor drainage and are periodically or permanently water saturated, and are therefore very different from the soils of the Planalto region.

Concerning land use, the Planalto region is predominantly used for pasture and agriculture, and the Pantanal, as it is a preservation area, has a greater number of natural areas than the Planalto. The most common land use classes in this region are Arborized Savanna and Forested Savanna in parts without frequent flooding, and Vegetation with River Influence and Humid Stepic Savanna in flooded regions.

The elevation also has very distinct characteristics in the Planalto versus the Pantanal, with elevated lands ranging from 200 to 1155 m in the Planalto, while the Pantanal presents flatter lands, with elevations below 200 m.

Table 3 shows the descriptive statistics of the quantitative variables – turbidity, discharge, rainfall and EWMA of rainfall – and the calculated Pearson correlation coefficient with SSC. The turbidity presented the best positive correlation with SSC ($R = 0.69$, $p < 0.05$), which was already expected due to the intrinsic relationship between this variable and the presence of sediments in the rivers. The discharge showed a negative correlation with SSC ($R = -0.17$, $p < 0.05$). Although it is a weak negative linear correlation, this value indicates that in the stations upstream of the basin (Planalto), where flows are lower and slopes are steeper, the concentration of suspended sediments is larger, whereas in the Pantanal region, where flows are higher and the flat land favours deposition, the SSC magnitudes are smaller.

The effect of past rainfall is observed in the median EWMA. For the daily rainfall, more than 50% of the rainfall data is zero (median equal to zero), but for the EWMA of rainfall it is above zero. Even if rainfall is zero on the day of or the day before SSC collection, the EWMA of rainfall aggregates information that occurred in longer previous periods. The EWMA of rainfall provides an increase in the linear correlation coefficient compared to the original rainfall data, which shows that this technique provides additional information for the SSC estimate.

Descriptive statistics and linear correlations between the SSC and turbidity at each station are presented in Table 4. Both SSC and turbidity presented higher values at stations located on the Planalto than at stations located in Pantanal (for minimum, medium and maximum values). This difference occurs because sediment production processes are intensified in the Planalto region due to the land use, soil type and steeper slopes. In the Pantanal, the extremely flat topography favours deposition processes, resulting in lower SSCs.

Regarding linear correlations, Planalto stations present the strongest positive linear relationships between SSC and turbidity, while linear correlations are lower in Pantanal stations, and in some cases negative. Sari *et al.* (2015) observed a negative linear correlation between turbidity and SSC for concentrations below 50 mg/L, i.e. lower values of turbidity were associated with higher concentrations of sediments, which does not reflect reality. According to the authors, linear adjustments are limited to concentrations equal to or greater than 50 mg/L. For smaller values, second-degree polynomial functions would be necessary to determine the SSC through turbidity.

According to Sari *et al.* (2015), in basins characterized by spatial variability in geology, soils and land cover, the characteristics of sediments that reach the rivers are heterogeneous, representing a mixture of materials of different origins and resulting in more complex relationships. However, if the properties of the sediment particles are constant, the relationship between turbidity and SSC may be linear.

These characteristics are observed at all 23 analysed stations. The stations located in the upstream parts of the basin show stronger linear relationships between SSC and turbidity,

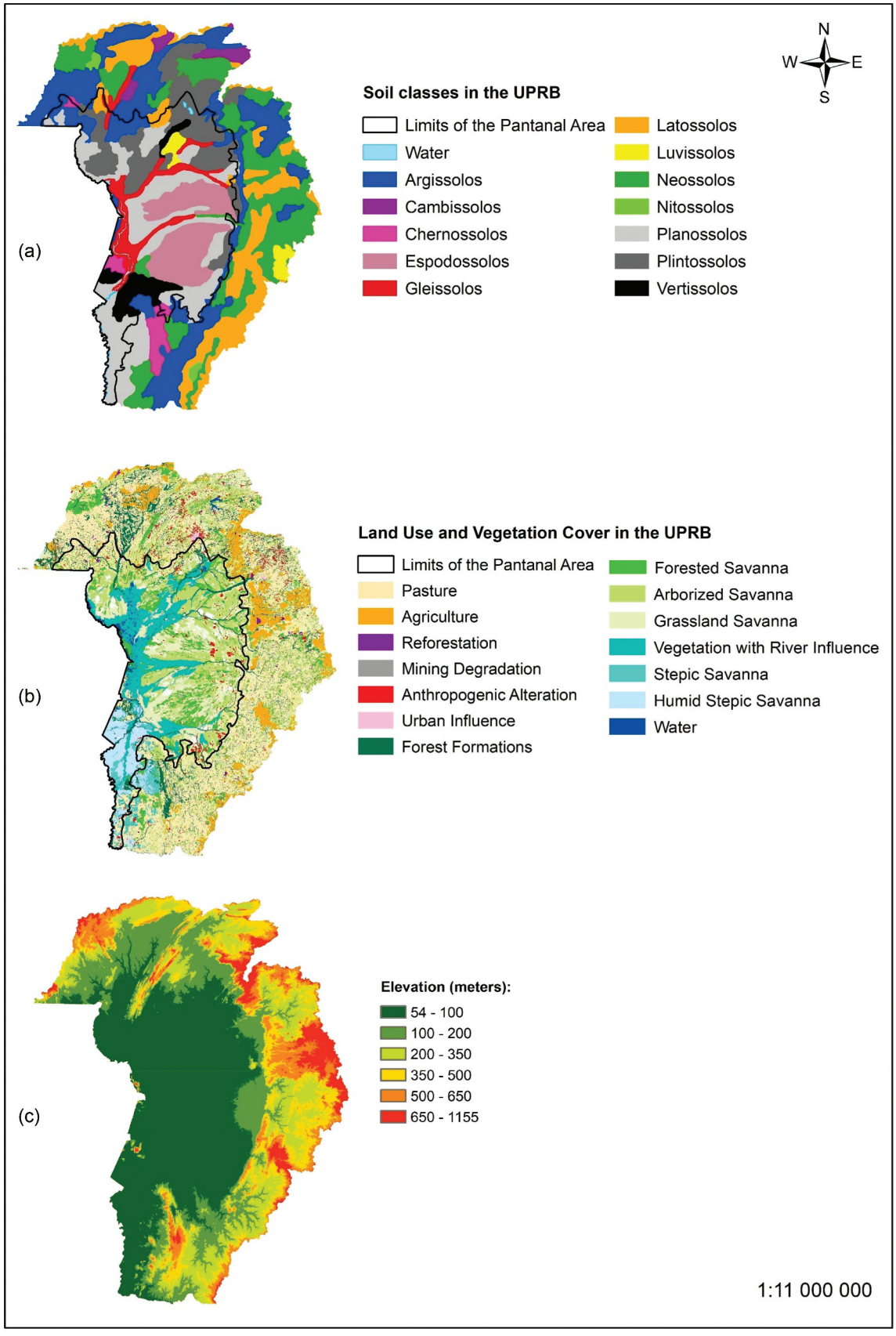


Figure 4. (a) Soil type classes, (b) land use classes and (c) digital elevation model in the Upper Paraguay River Basin (UPRB).

while the stations downstream, in the Pantanal, have lower linear correlation coefficients, indicating relationships of more complex degrees. This could be due to the great spatial variability of soil and land use characteristics and the flat topography in the Pantanal, which favours deposition, leading to low turbidity and SSC.

Table 3. Descriptive statistics of quantitative variables used in the Artificial Neural Network models.

	Model input and output variables	Min	Max	Mean	Median	Standard deviation	Linear correlation with Suspended Sediment Concentration
1	Turbidity (NTU)	0	1262	76.56	37.5	126.14	0.69
2	Discharge (m^3/s)	9.08	2488.89	370.96	214.82	454.71	−0.17
3	Daily rainfall t (mm)	0	109.47	3.13	0	9.92	0.06*
4	Daily rainfall t − 1 (mm)	0	119.42	3.77	0	10.52	0.16
5	Daily rainfall t − 2 (mm)	0	102.18	5.18	0	9.36	0.17
6	Daily rainfall t − 3 (mm)	0	121.44	4.02	0	9.98	0.14
7	Daily rainfall t − 4 (mm)	0	115.36	4.13	0	10.02	0.12
8	Daily rainfall t − 5 (mm)	0	108.98	4.55	0	10.11	0.08*
9	Exponentially Weighted Moving Averages of past rainfall h = 1 (mm)	0	55.23	3.03	0.79	5.79	0.19
10	Exponentially Weighted Moving Averages of past rainfall h = 2 (mm)	0	33.61	3.08	1.67	4.29	0.31
11	Exponentially Weighted Moving Averages of past rainfall h = 5 (mm)	0	17.08	3.05	2.37	3.02	0.37
12	Exponentially Weighted Moving Averages of past rainfall h = 10 (mm)	0	13.76	3.04	2.55	2.46	0.33
13	Exponentially Weighted Moving Averages of past rainfall h = 15 (mm)	0	12.07	3.07	2.68	2.27	0.22
14	Suspended sediment concentration (mg/L)	0	1736	128.68	51.45	200.59	1

* p value > 0.05.

Table 4. Descriptive statistics and linear correlations between Suspended Sediment Concentration and turbidity for each station.

	Station name	Region	Drainage area (km^2)	Suspended Sediment Concentration (mg/L)			Turbidity (NTU)			Linear correlation
				Min	Max	Mean	Min	Max	Mean	
1	São Pedro da Cipa	Planalto	3267	6.7	1736.0	309.8	4.4	1262.0	196.5	0.91
2	Rondonópolis	Planalto	12 297	19.4	593.2	259.4	14.4	770.0	229.2	0.76
3	Estrada BR-163	Planalto	3969	2.0	162.7	23.9	4.4	146.0	22.0	0.72
4	Garimpagem	Planalto	6328	98.1	1327.0	475.5	0.0	454.0	106.5	0.88
5	Coxim	Planalto	27 573	11.4	844.0	391.6	0.0	314.5	73.9	0.76
6	Estrada MT-738	Planalto	11 552	12.6	687.0	133.3	8.0	367.0	83.8	0.76
7	Miranda	Planalto	15 189	41.7	210.0	90.6	0.0	148.0	58.0	0.72
8	Ponte do Grego	Planalto	6318	15.0	284.9	120.2	2.5	275.0	67.7	0.85
9	Aquidauana	Planalto	15 707	19.3	337.6	116.7	15.9	297.0	76.6	0.81
10	Cáceres	Pantanal	32 373	9.5	83.5	33.1	0.0	74.2	34.5	0.80
11	Porto Cercado	Pantanal	37 024	27.5	182.1	59.3	12.1	218.0	61.9	0.66
12	Acima do Córego Grande	Pantanal	23 105	39.5	481.0	163.3	33.7	489.0	154.7	0.95
13	Descalvados	Pantanal	48 179	11.5	77.1	32.5	0.0	109.0	29.6	0.30
14	Porto Conceição	Pantanal	65 879	6.6	71.0	32.4	0.0	79.4	25.9	0.43
15	Ilha Camargo	Pantanal	39 006	16.1	156.2	64.7	10.4	224.0	61.1	0.59
16	São José do Boriréu	Pantanal	24 844	76.0	432.0	201.2	2.5	368.9	120.6	0.67
17	São Jerônimo	Pantanal	23 315	3.7	179.0	31.0	5.5	85.1	21.5	−0.16
18	São José do Piquiri	Pantanal	29 388	4.0	180.0	31.0	8.7	25.5	17.9	−0.20
19	Porto do Alegre	Pantanal	102 580	24.6	110 1	48.6	10.3	224.6	63.9	0.47
20	Amolar	Pantanal	72 642	0.0	90.5	34.9	10.9	109.0	36.0	0.34
21	São Francisco	Pantanal	185 113	0.0	57.0	28.1	0.0	94.6	30.2	0.38
22	Porto da Manga	Pantanal	261 480	7.3	88.7	38.9	0.0	131.0	34.1	0.26
23	Porto Ciriaco	Pantanal	17 190	20.7	487.0	104.4	0.0	144.0	55.0	0.33

The complex hydrological system and the great heterogeneity in the UPRB support the choice of a spatial approach to estimate SSC within the rivers of the basin, with the use of qualitative variables to characterize spatial variations and quantitative variables to characterize temporal variations, as well as the use of ANN, which are models with high capacity to approximate complex and nonlinear relationships.

5.2 ANN models for estimation of SSC in the UPRB

Thirteen regional ANN-based models were developed to estimate SSCs in the UPRB. Using the complete model, other models were tested by leaving variables or groups of variables out and evaluating the final performance of the ANN, to establish an understanding of the importance of each variable or group of variables and to identify the related physical-conceptual causes. The main discussion concerns these causes, and how they are best represented to model the process, with the purpose of extrapolating to unsampled locations.

Table 5 shows the performance statistics for the ANN models developed to estimate the SSC in the UPRB. The NS efficiency coefficient ranged from 0.499 to 0.943 during training and from 0.382 to 0.854 during verification. We observed that the other performance parameters, constituted by error statistics, always presented values consistent with those of NS, with smaller errors for larger NS, and larger errors for smaller NS. A number of the models were considered at least "good," according to the classification of Moriasi *et al.* (2015) for sediment models (NS > 0.70), both in training and in verification.

The first model (M01), in which all input variables were included, resulted in the best performance among all the performed tests, with NS score of 0.943 during training and 0.854

Table 5. Statistical performance of Artificial Neural Network models.

Model	M01	M02	M03	M04	M05	M06	M07	M08	M09	M10	M11	M12	M13
nni	5	4	4	4	4	3	4	4	3	4	3	3	3
Variables excluded	None	Turbidity	Discharge	Rainfall	EWMA of rainfall	Rainfall and EWMA	Drainage area	Average slope	Soil type	Land use	Soil type and land use	All, except turbidity	All, except turbidity and discharge
Training													
Ea(mean)	39.29	72.98	51.10	51.41	54.16	57.23	52.58	51.77	49.88	50.93	63.16	87.70	84.66
Ea(50)	19.57	32.05	27.47	27.58	31.41	33.88	26.92	26.47	26.58	27.46	33.44	53.12	52.25
Ea(95)	139.17	299.16	183.53	182.01	185.16	188.48	177.23	181.25	168.79	192.54	193.48	258.33	258.47
Ea(max)	265.13	1330.54	543.38	535.59	607.95	626.15	537.85	540.59	584.05	536.88	621.93	876.98	863.13
NS	0.943	0.499	0.822	0.825	0.774	0.742	0.813	0.818	0.828	0.819	0.748	0.511	0.519
Verification													
Ea(mean)	41.21	78.14	69.70	49.24	67.29	69.16	51.01	50.84	49.20	49.91	63.99	81.55	79.96
Ea(50)	23.81	32.00	46.77	29.96	48.56	50.17	30.03	30.56	33.98	29.73	44.07	54.83	59.25
Ea(95)	158.74	278.31	233.01	181.96	231.17	240.59	187.54	182.08	148.03	182.92	222.62	284.88	268.93
Ea(max)	250.06	406.87	477.57	258.09	446.13	498.57	272.40	257.75	341.63	283.43	468.31	597.17	582.02
NS	0.854	0.419	0.612	0.781	0.689	0.512	0.783	0.781	0.772	0.774	0.628	0.382	0.404

mean absolute error – Ea (mean)
maximum absolute error – Ea (max)
median absolute error – Ea (50)
95th percentile of the absolute errors – Ea (95)
Nash-Sutcliffe or NS efficiency coefficient – NS
Number of neurons in inner layer – nni

during verification ("very good," according to Moriasi *et al.* 2015) and lower error statistics (E_a(mean) of 39.29 mg/L during training and 41.21 mg/L during verification) in comparison with the other models. Figure 5 presents the distribution of errors and the approximation of observed and simulated data for this model in training and verification.

The M02 model, in which turbidity was left out, presented a considerable reduction in performance (NS = 0.419 and E_a (mean) = 78.14 mg/L during verification, therefore "not satisfactory") compared to M01 (and also compared to most of the other models), showing that turbidity is a crucial variable for the main model. However, in the simplest models, in which only turbidity was used as an input (M12), or turbidity and discharge (M13), the performance for the verification sample was also "not satisfactory" (NS = 0.382 and E_a(mean) = 81.55 mg/L). This confirms that the accuracy of the model for heterogeneous regions depends on the combination of several variables, and not only on turbidity.

The M03 model showed a reduction in performance statistics (only "satisfactory" for the verification sample, with NS = 0.612 and E_a(mean) = 69.70 mg/L), lower than that of the M02 but also quite significant, which means that discharge is also important for the models.

The model in which the EWMA of past rainfall (M05) was excluded produced worse results than the model in which daily rainfall data were excluded (M04). An NS coefficient of 0.689 and E_a(mean) of 67.29 mg/L were observed for M05 (only "satisfactory"), and an NS coefficient of 0.781 and E_a(mean) of 49.24 mg/L were obtained for M04 ("good"), both with the verification sample.

The model that did not include any rainfall data (M06) produced even greater errors and an NS coefficient of 0.512 with the verification sample, discouraging its use. This result shows that the rain data, mainly the transformations using the exponential moving average, are fundamental to obtaining more accurate SSC estimates in the UPRB.

In models M07 to M10, qualitative variables were excluded one at a time. Each model demonstrated a small performance reduction in its error statistics and NS coefficient, in relation to the complete model (M01), but still showed a "good" performance (NS > 0.75 in training and verification). However, when more than one qualitative variable was left out, the reduction in model performance with the verification sample became substantial, as observed in model M11 (NS = 0.628 and E_a(mean) = 63.99 mg/L), which excluded soil type, land use and land cover and which was classified as only "satisfactory." These results also confirm the importance of the qualitative variables.

6 Conclusions

We investigated the capacity of regional models to extrapolate SSC to ungauged basins in a heterogeneous region, using quantitative variables (turbidity, flow, precipitation and EWMA of past rainfall) to represent temporal variability in each basin and using qualitative variables (drainage area, soil type, land use, land cover and mean catchment slope) to represent spatial variability. Using ANN-based regional models, applied to the Brazilian part of the Upper Paraguay River Basin (362 380 km^2), we provided unprecedented answers to questions regarding not only the possibility to extrapolate SSC but also the relative importance of the variables for modelling.

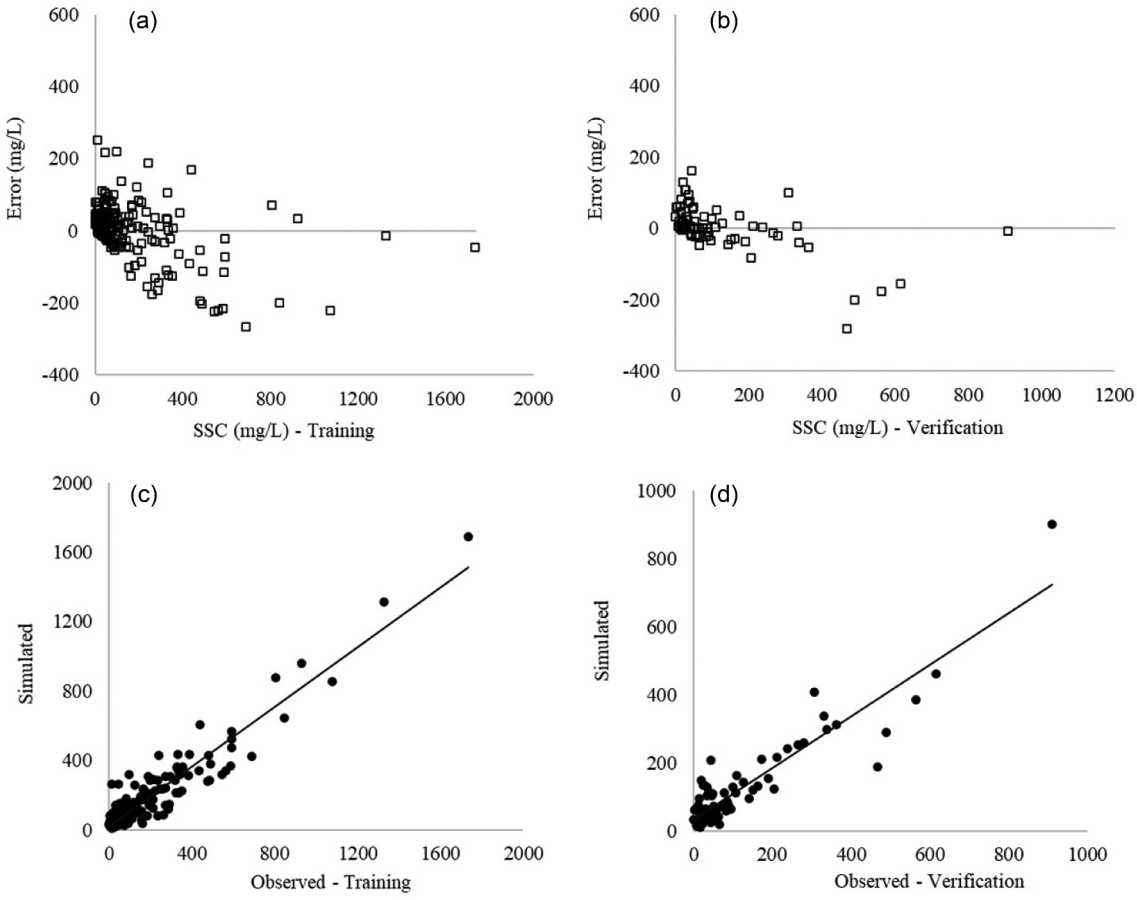


Figure 5. Verified error between the calculated and observed suspended sediment concentration (SSC) values (a, b) and the observed and calculated values in relation to the ideal fit (c, d), from model M01.

A complete model, with all qualitative and quantitative variables, proved to be adequate to predict SSC, achieving a "very good" performance classification. However, models using only turbidity and discharge proved unsuitable for extrapolation to other basins, resulting in "not satisfactory" performance. On the other hand, the exclusion (relative to the complete model) of only one of these variables resulted in "not satisfactory" and "satisfactory" classifications, respectively, which confirmed their importance when used in conjunction with the other variables. The exclusion of rainfall, in turn, especially the variable EWMA of past rainfall, resulted in models with only "satisfactory" classification, confirming the importance of precipitation, which is ultimately the main agent of hydro-sedimentological processes.

Regarding the importance of the qualitative variables, the models were still capable of achieving "good" performances when any of these variables were excluded individually. However, "good" classifications were not achieved if soil type, land use and land cover were excluded simultaneously.

We have shown that it is possible to estimate SSC in ungauged heterogeneous basins with "very good" performance, as long as turbidity and flow measurements are available, in combination with data from satellite products, including spatially varied data (drainage area, soil type, land use and mean catchment slope) and rainfall data.

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